

4.6 Detecting ranges with gaps ⚙️ ^

Settings

Beta features

PARTICIPATION
ACTIVITY

4.6.1: Using multi-branch if-else for detecting ranges with gaps: Movie ticket price

Start

☐ 2x speed

```
movie_ticket_price = None

user_age = int(input("Enter your age: "))

if user_age <= 12:      # Age 12 and under
    print("Child ticket discount.")
    movie_ticket_price = 11
elif user_age >= 65:    # Age 65 and older
    print("Senior ticket discount.")
    movie_ticket_price = 12
else:                  # All other ages
    movie_ticket_price = 14

print(f"Movie ticket price: ${movie_ticket_price}")
```

Memory	
95	
96	19
97	14
98	

Enter your
Senior ticket price
Movie ticket price

Enter your
Movie ticket price

Captions ^

1. After the user enters their age, the else-if branch's first branch checks if age is ≤ 12 .
2. user_age is 67, which is greater than 12, so the program moves to the second branch that checks if age is ≥ 65 .
3. 67 is ≥ 65 , so the second branch's statements execute, applying the senior discount to the ticket price and the program concludes by outputting the ticket price.
4. If the user's age falls between the gap of 12 and 65 (13 to 64), the else branch executes and applies the standard, more expensive price.

**PARTICIPATION
ACTIVITY**

4.6.3: Explicit ranges with gaps detection using logical AND and OR.

Start

2x speed

```
if office_num >= 100 and office_num <= 150:
    # valid office number
```

```
elif office_num >= 200 and office_num <= 250:
    # valid office number
```

```
else:
    # invalid office number
```

```
if (office_num >= 100 and office_num <= 150) or (office_num >= 200 and office_num <= 250):
    # valid office number
```

```
else:
    # invalid office number
```

Captions ^

1. The logical AND operator is used to identify the lower and upper bounds of the two valid ranges (100 to 150 and 200 to 250). Any number outside of the ranges is in the gap.
2. Further, the two ranges can be combined into a single branch using the logical OR operator.

**CHALLENGE
ACTIVITY**

4.6.1: Enter the output of the branch expressions.

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Start

Type the program's output

```
x = 4

if (x < 8) and (x > 3):
    print("a")
else:
    print("b")
```

a

1

2

3

4

5

Check**Next**View solution ▼ *(Instructors only)***CHALLENGE
ACTIVITY**

4.6.2: Ranges with gaps.

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Start

Integer `orange_count` is read from input representing the number of oranges. Output "Inadequate"

- the number of oranges is fewer than or equal to 10.
- or the number of oranges is more than 20.

[Click here for example](#) ▼

```
1 orange_count = int(input())  
2  
3 """ Your code goes here """  
4
```

1

2

3

Check**Next level**

View solution ▼ *(Instructors only)*